

House Price Prediction Using Machine Learning

Project Overview

The objective of this project was to develop a machine learning model capable of predicting house prices based on various property characteristics. The dataset used for this analysis contained information such as area, number of bedrooms, bathrooms, stories, parking spaces, air conditioning availability, and furnishing status. The project aimed to identify the key factors influencing house prices and compare the performance of different regression models.

Data Preparation and Processing

The dataset was loaded using Pandas and examined for missing values and duplicate records. Duplicate entries were removed, and categorical features were converted into numerical format using One-Hot Encoding. The hot water heating feature was excluded from the analysis due to its low impact on house prices. After preprocessing, the data was split into training and testing sets using an 80:20 ratio.

Model Development

Two machine learning algorithms were implemented and evaluated:

1. Linear Regression
2. Random Forest Regressor

The models were trained using the processed dataset and evaluated using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R^2 Score. The Random Forest Regressor outperformed Linear Regression by capturing more complex relationships between property features and house prices.

Key Findings

The analysis revealed that area, number of bathrooms, bedrooms, parking spaces, and air conditioning were the most influential factors affecting house prices. Larger houses with additional amenities generally commanded higher prices. A strong positive correlation was observed between house area and selling price. It was also interesting to note that location-related features and modern amenities significantly contributed to property value.

Conclusion

The developed machine learning model successfully predicted house prices with good accuracy. Among the tested models, Random Forest Regressor produced the best results and demonstrated superior predictive capability. Based on the findings, real estate businesses should focus on promoting larger properties with modern amenities and favourable locations, as these factors have the greatest impact on market value. This project demonstrates the practical application of machine learning techniques in real estate price estimation and decision-making.